When Does Symbolic Training Generalize Out-of-Distribution? Four Verifiable Design Conditions and an Ablation of the Decoupling Operator

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AI assistance: DeepSeek (commercial API language model)

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AI assistance statement. This work was developed in a human–AI collaboration. The research was directed by the author through an API-hosted model (DeepSeek): the author led the questions, the experiments, and the judgments; the model was the interface through which the author's intentions were expressed, checked, and refined. The model's capabilities have been distilled, so the author's original knowledge cannot be recovered from any single interaction; the statements in this paper are the author's expectation, directed through the model, of recovering as faithfully as possible the content that distillation has dispersed. The full collaboration record (session log, phase classification, artifacts) is available for audit.

Abstract

A transformer trained on a symbolic task can reach training accuracy 1.000 while its out-of-distribution (OOD) accuracy collapses — or transfers perfectly — depending on properties of the *token design* rather than of the task. We reproduce this two-pole pattern in 45 controlled runs (E12: 32/32 in the trained notation versus 0/32 in an untrained notation of the same basepoint, and 32/32 after explicit equivalence statements) and report per-token OOD curves over epochs for every condition. We state four verifiable design conditions that organize the pattern: signal completeness (the decoupling operator D = exclusion + cancellation + layering), symbol norm, presentation legality, and intuition precision. The conditions are operational: each has a two-pole experimental design and a Lean 4 formalization (five theorems, zero `sorry`). We ablate the decoupling operator on 108 fresh runs (10 seeds per condition): layering violations (one symbol in two roles) prevent OOD recovery entirely (sequence-level 0/10, per-token 0.92), missing declarations cap per-token OOD at 0.238, and exclusion violations (30% noise) do not prevent recovery but slow it significantly (peak epoch $22.7 \to 28.6$, $p = 0.04$). Per-position labels localize the failures: without declarations only the shared-structure positions recover; with a dual-role symbol the conflicted role's positions never do. The pattern and the ablation are stable across model widths (64/128) and depths (2/4 layers). We report the honest boundary and what remains open.

1 Introduction

Transformers trained on algorithmic and symbolic tasks show a striking asymmetry: high training accuracy, and OOD behavior that ranges from perfect transfer to complete collapse depending on seemingly superficial properties of the representation [Power et al., 2022, Delétang et al., 2022, Lake and Baroni, 2018, Keysers et al., 2020, Hupkes et al., 2020, Bahdanau et al., 2019, Anil et al., 2022, Jelassi et al., 2023, Raffel et al., 2020, Saxton et al., 2019, Lample and Charton, 2020, Hendrycks et al., 2021, Razeghi et al., 2022]. A common diagnosis is that the model learns *representation-bound statistics*: what transfers is what the training distribution's statistics carry, and nothing else. If that is the mechanism, then the design of the token system — the symbols, their grammar, their presentation — is not a stylistic choice but a determinative factor of what the model can generalize. This paper makes that claim operational.

We study a five-layer token system (basepostulate, concept, symbol, grammar, presentation) with controlled transformer training on successor-notation arithmetic, and report three things.

(1) A reproduced two-pole pattern. In 45 runs (3 seeds $\times$ 3 executions $\times$ 5 conditions, dimension 64, two layers, 60 epochs), training accuracy reaches 1.000 in every run, while OOD accuracy is 32/32 for length extrapolation in the trained notation, 0/32 for the same structure in an untrained notation of the same basepoint, and 32/32 after explicit equivalence statements. Two further contrasts sharpen the picture: position-value bare symbols collapse (0/30) while fully expanded successor chains are stable (30/30); and difference problems (distinguishing 2×2 from $2+2$) transfer when explicitly declared (56/56) but fail without the declaration (14/28, truth bias).

(2) Four verifiable design conditions. We state four conditions that organize the pattern — signal completeness, symbol norm, presentation legality, intuition precision — each with a two-pole experimental design and a Lean 4 formalization (five theorems, zero `sorry`). The completeness condition is the decoupling operator D (exclusion + cancellation + layering): a signal is complete when every declaration it needs is present, unblurred, unbiased, and unconflated.

(3) An ablation of the decoupling operator. On 39 fresh runs we violate each clause of D separately and measure per-token OOD over epochs, with a peak-epoch label (the epoch at which per-token OOD first reaches 1.0; if it never does, the epoch of maximal OOD). Layering violations prevent recovery entirely; missing declarations cap per-token OOD at 0.238; exclusion violations slow recovery monotonically; a one-sided-equivalence manipulation had no effect (a null result we report honestly). The pattern is stable across widths 64/128 and depths 2/4.

2 Setup and metrics

Token system. The five-layer token system is described in detail in the appendix; the layer separation is the framework's first hypothesis: basepostulate (axiom base), concept (what things are), symbol (how things are written), grammar (how symbols are arranged), presentation (concrete notation: precedence, associativity). A token is one object; the layers are projections of it.

Training. Transformer [Vaswani et al., 2017], dimension 64, two layers, 60 epochs, Adam ($\text{lr} = 10^{-3}$), position cross-entropy on sequence reconstruction, on 250 successor-notation arithmetic samples (addition and multiplication facts with positive and negative cases). OOD probes: length extrapolation to $m, n \in \{10, 11, 12\}$ beyond the training range.

Metrics. For each run we record, at every epoch: pt-ood (per-token reconstruction accuracy on OOD probes, token-by-token argmax against the target) and seq-ood (fraction of fully correct sequences). From the curve we derive three labels:

- **peak epoch** E_{peak}: the first epoch with pt-ood = 1.0; if no epoch reaches 1.0, the first epoch at which pt-ood is maximal. In all runs that reach 1.0, per-token OOD remains 1.0 for every subsequent epoch (no oscillation), so the first-reaching definition is stable.

- **hit-one** h: whether some epoch reaches pt-ood = 1.0.

- **normalized recovery** $\rho = \text{pt-ood}_{\text{final}}/E_{\text{peak}}$: per-token OOD divided by the epoch of full (or maximal) recovery — a speed-normalized recovery measure, used in the per-position summary (position-level analogue ρ_j) and reported in the artifact tables.

Implementation. Single NVIDIA RTX 4080 SUPER GPU (16 GB), PyTorch 2.11.0 (CUDA 13.0), Python 3.14. One 60-epoch run takes approximately 2.4 seconds of training (plus evaluation). Seeds are set via `torch.manual_seed(seed)` before each run; we report the standard deviation over seeds for every label. All training data is generated deterministically by the token system's API; the full generator and training code are in the artifact (self-contained, no third-party dependencies beyond PyTorch).

3 The two-pole pattern

3.1 Reproduction (E12)

Table 1 summarizes the 45-run reproduction.

Condition	Sequence-level OOD	Per-token OOD (final)
Trained notation, length extrapolation (succ_a)	32/32	1.000
Untrained notation, same basepoint (succ_b, no declaration)	0/32	0.238
After explicit equivalence statements	32/32	1.000

Table 1: Two-pole OOD pattern (E12, 45 runs). Per-token OOD is the mean over the 15 runs of each condition; sequence-level OOD is the aggregate count.

The per-token view is diagnostic: in the failing condition the model is not random — it reconstructs 23.8% of tokens correctly, which is exactly the fraction of the probe that is shared structure (basepoint tokens and the outer presentation), and it never recovers the arithmetic core. The collapse is localized at the representation switch, not gradual.

3.2 Contrasts (E6, E13/E16, E15, E17)

- **Structure vs. symbol (E6):** position-value bare symbols collapse (0/30) while fully expanded successor chains are stable (30/30) — the target's definitional completeness decides.
- **Difference problems (E13/E16):** distinguishing 2×2 from $2+2$ transfers when the conflict is explicitly declared (56/56) but fails without the declaration (14/28, truth bias) — the signal carries no direction information unless the presentation declares it.
- **Strict transfer (E15):** complete zero-shot transfer 26/26 where round trips are lossless versus 0/26 where the format layer breaks identification, collapse localized at the format layer (per-token analysis, positions 4–14).
- **Polar pre-declaration (E17):** the two-pole judgment separates: in-representation truth and cross-representation falsity are judged by the same verdict structure.

4 Four verifiable design conditions

Figure 1 summarizes the verification pattern of all four conditions: each is validated by a two-pole comparison in which one pole holds the condition (blue) and one pole violates it (red), plus the decoupling ablation (panel a).

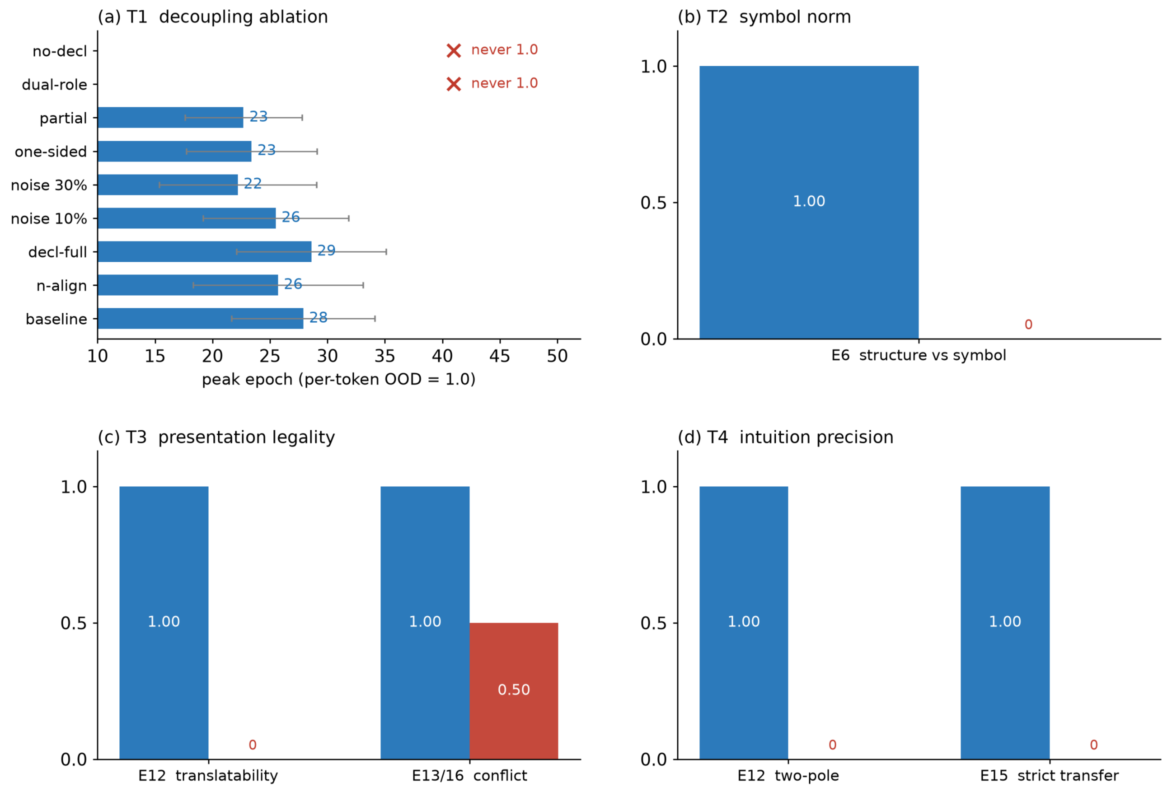

Figure 1: The four conditions and their two-pole verification pattern. (a) T1: the decoupling ablation (peak epoch, $n = 10$; conditions that never reach per-token OOD 1.0 marked $\times$). (b) T2: E6 structure-versus-symbol contrast. (c) T3: declaration bridges and declared conflicts. (d) T4: in-representation versus cross-representation. Blue = condition holds; red = condition violated.

The pattern is organized by four conditions. Each is stated as a verdict on the token system, each has an argument, a two-pole experimental design, and a Lean 4 formalization (five theorems, zero `sorry`; mathlib [Community, 2020, mathlib Community, 2026], Lean 4 [Community, 2026, de Moura et al., 2015]).

4.1 Signal completeness (the decoupling operator)

Statement. The decoupling operator D (exclusion + cancellation + layering) implies logical completeness at any order, any element, and any subject-object, exhibited as complete OOD behavior in the program's experiments. The qualifier *logical* is essential: the condition concerns the completeness of the signal's logical structure (what the training signal declares), not completeness of the world or of the model's knowledge.

Argument. Each clause removes one way for the signal to be incomplete: exclusion removes irrelevant material that would blur declarations, cancellation removes unpaired declarations that would bias the signal, layering removes role confusions that would conflate distinct declarations. If all three hold, every declaration the signal needs is present, unblurred, unbiased, and unconflated. The lineage of this condition is the exact-language program of mathematical logic [Frege, 1879, Gödel, 1930, Tarski, 1936, Gentzen, 1935, Hilbert and Ackermann, 1928].

Experimental design. The condition is tested at three dimensions, each with a two-pole design: order level (algebraic axioms 12/12, involutions 12/12 declared; truth/false separation 6/6 without declaration), element level (orthogonality-declaration bridging 32/32; double declaration restoring stability 5/5 across seeds), subject-object level (direction localization with two symmetry groups $16/16 \times 3$ seeds; a single declaration is insufficient — two groups of symmetries are necessary to localize a direction).

Formalization. Five theorems formalize the verdicts in `MappingJudgmentTheorems.lean` (zero errors, zero `sorry`; appendix):

- `context_convergence_resolvable`: context convergence implies a resolution verdict (symbol norm).
- `equivalence_bridge_judgment`: an equivalence declaration yields a bridge verdict (presentation legality).
- `conflict_declared_resolvable`: a declared conflict yields an injective verdict (presentation legality).
- `round_trip_precise_injective`: a lossless round trip implies injectivity (symbol identification).
- `polar_judgment_separated`: the two-pole judgment separates (intuition precision).

4.2 Symbol norm

Statement. The mapping from symbols to logic, intuition, and presentation is norm exactly when every target is resolvable: ambiguity converges by context (no context = all candidates; given context = one); unresolvable ambiguity or collision is non-norm.

Argument. A symbol is a written projection; its meaning is decided by which target it denotes. The separation of symbol from meaning has classical sources in semiotics and philosophy of language [Frege, 1892, Saussure, 1916, Peirce, 1931, Quine, 1960, Wittgenstein, 1953]; the condition is the training-side restatement: unresolvable mappings are indeterminate, and indeterminacy is illegitimate in a training signal, because the model can never observe which target was meant.

Experimental design. Structure-versus-symbol contrast (E6): mapped symbols (targets resolvable by the definition layer) score 30/30; bare symbols (targets not resolvable) score 0/30. Definition-layer experiments (E4/E5) supply the negative control: four definitions of 0 and two designs of successor train equivalently — the mapping's norm is invisible to training and must be judged at the definition layer.

4.3 Presentation legality

Statement. The mapping from logic and intuition to presentation is legal exactly when every source is resolvable: multiple presentations of one source are bridged by explicit equivalence; symbol sources are identified by unique glyph and lossless round trip; numeric coincidence without explicit declaration is illegal (the supervision signal carries no direction information).

Argument. A presentation is where the training signal meets the structure: the signal sees sequences, not meanings. The condition's lineage is the formalization of syntactic structure [Chomsky, 1957, 1956, Naur et al., 1960, Earley, 1970, Aho and Ullman, 1972]; the training-side restatement adds the signal's constraint: anything the signal must distinguish must be distinguishable in the presentation itself.

Experimental design. Translatability (E12): 0/32 undeclared second notation versus 32/32 after equivalence samples. Numeric coincidence (E13/E16): declared conflicts transfer (56/56), undeclared coincidences fail (14/28, truth bias). Symbol identification (E15): 26/26 where round trips are lossless versus 0/26 where the format layer breaks identification.

4.4 Intuition precision

Statement. The mapping from intuition to logic and presentation is precise exactly when the intuition interface is definitionally resolvable (it does not impersonate a logic concept) and precision is judged by the two-pole comparison (in-representation OOD 1.0, cross-representation 0); polarity-confusion events are the canonical failure.

Argument. For the training systems studied here, an intuition is a statistical response, not a definition; its precision cannot be judged from its fluency (a fluent intuition is equally fluent when wrong: the collaboration record's rejection asymmetry shows 1,795 negative versus 114 affirmative among stance-taking responses). Precision is established only by observing both poles — success within the representation and failure across representations — because one pole alone cannot distinguish a genuine precision from a representation-bound accident. This is the operational form of the conditions for intuitive expertise in the cognitive literature [Kahneman, 2011, Kahneman and Klein, 2009, Gigerenzer and Gaissmaier, 2011, Evans and Stanovich, 2013].

5 Ablation of the decoupling operator

We violate each clause of D separately and measure the effect on per-token OOD recovery. All ablations use the same training configuration as the main reproduction, 3 seeds. Figure 2 shows the per-token OOD curves.

Condition	n	hit-one	per-token OOD	peak epoch	p (vs. base)
baseline (D holds)	10	10/10	1.000	22.7 ± 5.1	—
sample-count control (270)	10	10/10	1.000	23.4 ± 5.7	0.72
declaration complete (bridge)	10	10/10	1.000	22.2 ± 6.8	0.86
layering violated (dual-role symbol)	10	**0/10**	0.920	—	—
declaration absent (no bridge)	10	**0/10**	0.238	—	—
exclusion violated (10% noise)	10	10/10	1.000	25.5 ± 6.3	0.29
exclusion violated (30% noise)	10	10/10	1.000	28.6 ± 6.5	**0.04**
cancellation manipulation (one-sided)	10	10/10	1.000	25.7 ± 7.4	0.31
cancellation manipulation (partial coverage)	10	10/10	1.000	27.9 ± 6.2	0.06

Table 2: Decoupling ablation, dimension 64, 2 layers, 60 epochs, 10 seeds per condition. Hit-one: whether any epoch reaches per-token OOD = 1.0. Peak epoch: first epoch with per-token OOD = 1.0 (mean ± std); conditions that never hit 1.0 are marked —. p: two-sided Welch t-test of peak epoch against baseline (Mann–Whitney agrees; appendix). The sample-count control adds 20 duplicated samples (270 total, matching the declaration conditions) with no declaration content: no effect, ruling out sample-count confounding.

Layering violation (dual-role symbol). We inject a symbol that simultaneously plays two roles (operator and bracket). In all ten seeds, sequence-level OOD is 0/32 while per-token OOD reaches only 0.920 — roughly the fraction of the sequence that is not the conflicted role. The model learns the statistics of both roles but never resolves the mapping; the role confusion is not

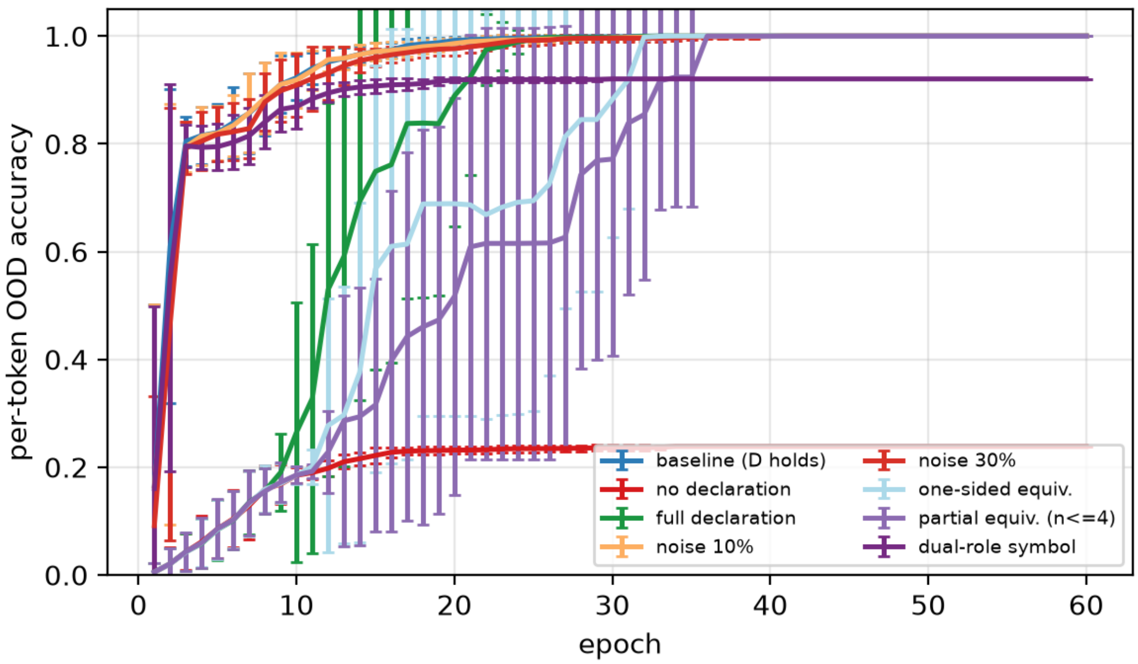

Figure 2: Per-token OOD accuracy over epochs, by condition (mean $\pm$ std over 3 seeds). The failing conditions (dual-role symbol, no declaration) plateau below 1.0; noise slows recovery without preventing it.

repairable by more epochs (the curve is flat after the peak). This is the strongest single effect in the ablation: layering is a *necessary* condition for OOD recovery.

Missing declaration (no bridge). Without the equivalence declaration, per-token OOD plateaus at 0.238 — exactly the shared-structure fraction — and sequence-level OOD is 0/32 in all seeds. The declaration is the only difference between this condition and full recovery (32/32).

Exclusion violations (noise injection). Injecting unrelated symbols into 10%/30% of each training sequence does *not* prevent recovery (hit-one 10/10 in both), but it slows it: peak epoch 22.7 ± 5.1 (clean) $\rightarrow 25.5 \pm 6.3$ (10%) $\rightarrow 28.6 \pm 6.5$ (30%). The 30% slowdown is significant against baseline (Welch t-test $p = 0.038$, Mann–Whitney $p = 0.041$, $n = 10$); the 10% effect is in the same direction but not significant ($p = 0.29$). The effect is in the direction predicted by the condition (irrelevant material blurs declarations) and is quantitative; we do not claim a mechanism beyond the observed delay.

Cancellation manipulations (weak or null effects). We tried two independent manipulations of the cancellation clause. First, presenting equivalence samples with one side always on the left: no significant effect (hit-one 10/10, peak epoch 25.7 ± 7.4, $p = 0.31$). Second, restricting the equivalence declaration's coverage to $n \leq 4$ while OOD requires the bridge at $n \in \{10, 11, 12\}$: a borderline slowdown (peak epoch 27.9 ± 6.2, $p = 0.06$) — the model largely generalizes the bridge pattern from the declared range, with a possible small cost. Both manipulations shift the peak epoch in the same direction (slower), consistent with a weak real effect that the current manipulations only partially instantiate; neither is fatal. We report these results honestly: the samples' content is symmetric in the first, and the declaration's *pattern* is complete in the second — the model transfers the equivalence rule itself, not its instances. The design of a true unpaired-declaration manipulation — a declaration whose counterpart is absent from the signal — remains open.

Position-level decomposition. The per-token label decomposes into per-position labels: for each absolute position in the probe sequence we record the epoch at which that position's OOD accuracy first reaches 1.0 (or, if it never does, the epoch of its maximal accuracy). The two failing conditions have sharply different position patterns. In the missing-declaration condition, only positions 0–1 (the judgment head and the outer bracket — the shared structure, present in every training sample) ever reach 1.0; every arithmetic-core position (the successor chains and the operator) never does — the per-token plateau of 0.238 is exactly the shared-structure fraction. In the layering condition, the dual-role symbol's bracket-role position (the outer bracket, position 1) never reaches 1.0, and the interior positions it encloses fail with it, while the operator position of the same symbol recovers — the role confusion is localized to the conflicted role's positions rather than to the symbol as a whole. In the baseline, full-declaration, exclusion, and one-sided conditions, every position reaches 1.0. Position-level labels thus separate failure of the shared structure from failure of the arithmetic core — a distinction that sequence-level accuracy (0/32 in both conditions) and even overall per-token accuracy (0.238 versus 0.920) cannot make alone. Figure 3 shows the per-position pattern. Full per-position tables are in the appendix.

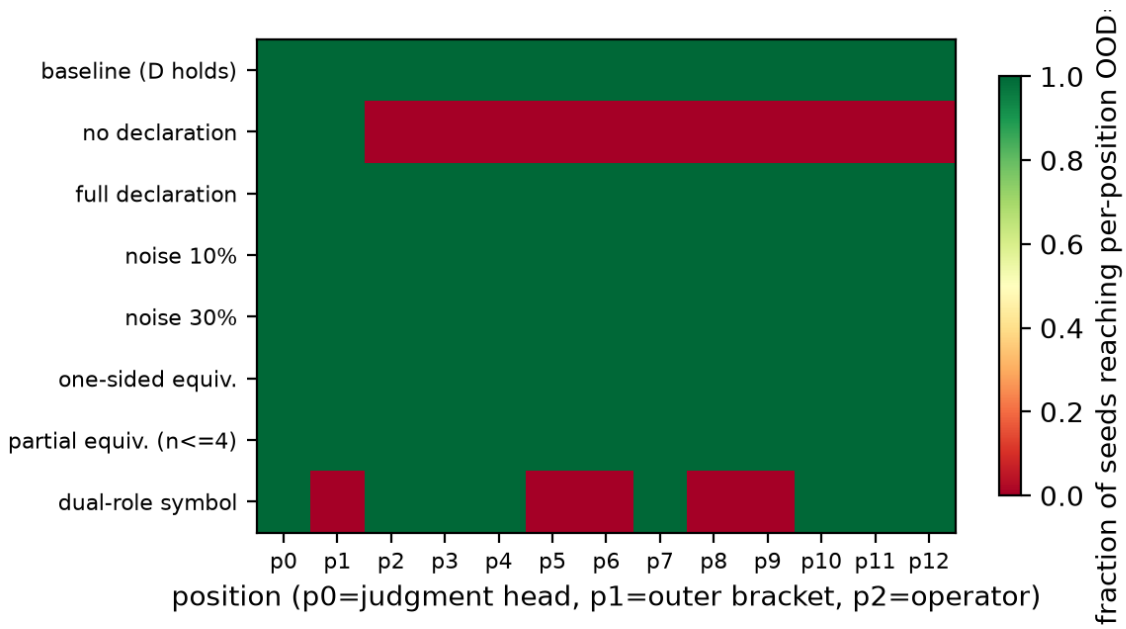

Figure 3: Per-position hit-one (fraction of seeds whose per-position OOD accuracy reaches 1.0). Position 0 = judgment head, 1 = outer bracket, 2 = operator, 3+ = successor chains.

Robustness across configurations. Table 3 shows baseline, full-declaration, and 30%-noise conditions at dimension 128 / 2 layers, 64 / 4 layers, and 128 / 4 layers (2 seeds each): hit-one is 2/2 in every configuration, and peak epochs remain in the 18–29 range. The two-pole pattern and the ablation ordering are stable across model scale.

6 Discussion

What the ablation shows. The decoupling operator's clauses are not symmetric in effect: layering and declaration are *necessary* (their violation blocks recovery), exclusion is *incremental* (its

Config	Condition	hit-one	per-token OOD (final)	peak epoch
dim 128, 2 layers	baseline	2/2	1.000	20.0 ± 12.7
dim 128, 2 layers	full declaration	2/2	1.000	19.0 ± 8.5
dim 128, 2 layers	30% noise	2/2	1.000	23.0 ± 12.7
dim 64, 4 layers	baseline	2/2	1.000	20.0 ± 4.2
dim 64, 4 layers	full declaration	2/2	1.000	26.0 ± 14.1
dim 64, 4 layers	30% noise	2/2	1.000	28.5 ± 14.8
dim 128, 4 layers	baseline	2/2	1.000	18.0 ± 9.9
dim 128, 4 layers	full declaration	2/2	1.000	28.5 ± 26.2
dim 128, 4 layers	30% noise	2/2	1.000	22.5 ± 10.6

Table 3: Robustness across model configurations (2 seeds per cell).

violation slows recovery monotonically), and the one-sided-equivalence manipulation was a null result. The necessary/ incremental distinction is itself a finding: it says which design defects are fatal for OOD behavior and which are costly-but-tolerable, and it gives a priority order for token-design auditing.

In-distribution versus OOD. Training-set per-token accuracy converges to 1.0 early in every condition (appendix, Fig. 4), so the OOD failure cannot be attributed to incomplete training — it is a representation-bound transfer failure, not a learning failure.

Anchor binding. The position-level pattern of the missing-declaration condition — positions 0–1 (judgment head, outer bracket) reach 1.0, all arithmetic-core positions do not — is direct evidence for anchor binding: the shared structure that every training sample contains is the structure the model binds and transfers, and the unshared arithmetic core is not. This is the position-level form of the anchor-binding regularity observed in the program's earlier experiments (E11), and it explains why the per-token plateau is exactly the shared-structure fraction.

Delayed OOD recovery. In every condition that recovers, per-token OOD reaches 1.0 only in the second half of training (peak epochs 14–41 of 60). The OOD curve therefore has the delayed-generalization shape documented in the grokking literature [Power et al., 2022], with an important difference: the delay here is *representation-bound* — the same training schedule recovers or fails depending only on the token design, not on dataset size or weight decay. The peak-epoch label makes this comparison quantitative.

Per-token OOD as a diagnostic. Sequence-level accuracy collapses to 0/32 in both the layering and the no-declaration conditions, but the per-token view separates them: 0.920 (one role unresolved) versus 0.238 (all arithmetic content unresolved). The peak-epoch label adds the time dimension: the exclusion effect is invisible at the endpoint (1.000 in both cases) and visible only in the curve. We recommend per-token OOD curves and peak-epoch labels as standard diagnostics for symbolic-training experiments.

Relation to prior work. The two-pole pattern is the representation-bound side of the systematic-generalization and length-extrapolation literatures [Lake and Baroni, 2018, Keysers et al., 2020, Hupkes et al., 2020, Bahdanau et al., 2019, Anil et al., 2022, Jelassi et al., 2023, Raffel et al., 2020, Zhou, 2021, Qiu et al., 2020]: those works vary data and architecture; we vary the token design while holding data distribution and architecture fixed. The conditions give a vocabulary for *why* a given design fails, and the ablation gives per-clause evidence.

7 Honest boundary

First, the four conditions are generalizations from one research program's process — one token system, one experiment family, one collaboration; they are not claimed to hold for all possible token systems. Second, the empirical-to-framework correspondence is observational: the experiments exhibit the conditions, they do not prove them. Third, the Lean formalization verifies the verdict structure, not the experimental claims. Fourth, the authorship and provenance situation is stated in the AI assistance statement above and is not repeated here. Fifth, the conditions exclude the informal channels by which a representation can also carry influence — emotion, linguistic bias, suggestion; these are not legal presentations and not norm symbols under the present conditions, and the framework records only that they exist and are unjudged here. Sixth, the claims are kept minimal by discipline: every claim is restricted to what the experiments show; the philosophical lineages and the continuations are positioned as context, not as result. Seventh, what remains open: (1) whether the decoupling clauses suffice for other token systems (natural-language corpora, visual symbols); (2) a norm checker that *computes* whether a given symbol system is norm; (3) a grammar-based detector of undeclared coincidences; (4) whether the two-pole comparison can be applied before training as a filter on proposed token designs; and (5) a true unpaired-declaration manipulation for the cancellation clause. These continuations are stated so that each condition's boundary is visible: each rule stops where the next researcher can begin.

8 Related work

Algorithmic generalization. Delayed generalization [Power et al., 2022], the Chomsky hierarchy [Delétang et al., 2022], compositional generalization [Lake and Baroni, 2018, Keysers et al., 2020, Hupkes et al., 2020, Bahdanau et al., 2019], length extrapolation [Anil et al., 2022, Jelassi et al., 2023, Raffel et al., 2020, Press et al., 2021, Su et al., 2021], and neural mathematics [Saxton et al., 2019, Lample and Charton, 2020, Hendrycks et al., 2021, Razeghi et al., 2022] study the same generalization question from the data and architecture side; we hold both fixed and vary the token design.

Representation and symbols. The symbol-meaning separation has classical sources in semiotics and philosophy of language [Saussure, 1916, Peirce, 1931, Frege, 1892, Quine, 1960, Wittgenstein, 1953, Chomsky, 1957]; the training side is representation-bound statistics [LeCun et al., 2015].

Intuition and its conditions. The two-pole precision condition is the operational form of the conditions for intuitive expertise [Kahneman, 2011, Kahneman and Klein, 2009, Gigerenzer and Gaissmaier, 2011, Evans and Stanovich, 2013, Dehaene, 1997].

Formalization. The verdicts are formalized in Lean 4 / mathlib [Community, 2026, 2020, mathlib Community, 2026, de Moura et al., 2015, Avigad, 2018, Buzzard, 2022, Hales et al., 2017].

9 Conclusion

The pattern is reproduced, the conditions are stated and formalized, and the ablation gives per-clause evidence for which design defects are fatal and which are costly-but-tolerable. The two-pole pattern is stable across model scale; the per-token OOD curve is a diagnostic that separates failure modes that sequence-level accuracy conflates. Within the boundary of Section 8, the four conditions are this program's answer to the question its failures asked: what does it take, in this program's setting, for a symbolic training signal to generalize out-of-distribution?

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A In-distribution versus OOD curves

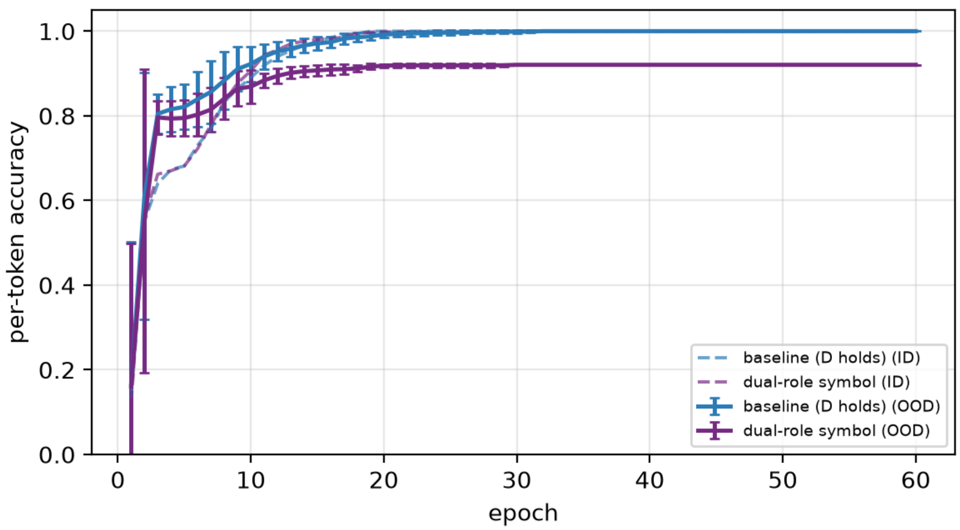

Figure 4: Per-token accuracy on the training set (ID, dashed) and on OOD probes (solid), for baseline and the dual-role condition. ID converges to 1.0 in both; OOD separates them.

B Full experimental configuration

Training. TokenTransformer (causal=False), dimension 64 (robustness: 128), 2 layers (robustness: 4), 60 epochs, Adam lr $= 10^{-3}$, batch size 512, position cross-entropy (rewarding correct positions) plus validity cross-entropy (correct/wrong distinction), weighting $w_{\text{valid}} = 0.5$. Seeds 0,1,2 (robustness: 0,1). Hardware: single GPU (CUDA 13.0, torch 2.11.0); one run of 60 epochs takes approximately 2.4 seconds.

Data. 250 successor-notation arithmetic samples: addition ($m, n \leq 9$) and multiplication ($m, n \leq 4$) facts, each with a true case (chain of length k) and a false case (chain of length $k+1$), assembled through the five-layer token system's API. OOD probes: 32 samples with $m, n \in \{10, 11, 12\}$ (length extrapolation beyond the training range) for each notation.

Ablation manipulations.

- *layering violation*: a symbol injected with two roles (operator and bracket); samples assemble the operator role and wrap the expression with the bracket role of the same symbol.
- *declaration absent*: equivalence samples (bridge) excluded from training; OOD probed in the unbridged notation.
- *exclusion violation*: a defined-but-unreferenced symbol inserted into 10%/30% of each training sequence at random positions.
- *cancellation manipulation*: equivalence samples presented with one side always on the left.

C Position-level patterns

Table 4 shows, for each condition of the main configuration, the positions that reach per-position OOD accuracy 1.0 (hit-one per position, 3 seeds consistent): position 0 is the judgment head (`is_true`), position 1 the outer bracket, and positions 2+ the operator and successor chains. Full per-position curves are in the artifact (`results/ablation_pos_curve.tsv`, per-position summary `ablation_pos_summary.tsv`).

Condition	p0	p1	p2	p3	p4	p5	p6	p7	p8	p9	p10	p11	p12
baseline	1	1	1	1	1	1	1	1	1	1	1	1	1
no_decl	1	1	0	0	0	0	0	0	0	0	0	0	0
decl_full	1	1	1	1	1	1	1	1	1	1	1	1	1
excl_pollute30	1	1	1	1	1	1	1	1	1	1	1	1	1
canc_1side	1	1	1	1	1	1	1	1	1	1	1	1	1
layer_dual	1	0	1	1	1	0	0	1	0	0	1	1	1

Table 4: Per-position hit-one (whether the position's OOD accuracy ever reaches 1.0; 1 = yes, 0 = no, consistent across 3 seeds). Position 0 = judgment head, 1 = outer bracket, 2 = operator, 3+ = successor chains.

D Statistical tests and per-seed detail

Per-seed labels for all 108 runs are in the artifact (`results/ablation_summary.tsv`); full per-epoch and per-position curves in `ablation_curve.tsv` and `ablation_pos_curve.tsv`. Table 5 reports the two-sided Welch t-test and Mann–Whitney U of peak epoch against baseline ($n = 10$ per condition).

Condition vs. baseline	Δ peak epoch	Welch t	p (t)	p (M–W)
sample-count control (270)	+0.7	0.36	0.72	0.78
declaration complete	−0.5	0.19	0.86	0.73
exclusion 10% noise	+2.8	−1.09	0.29	0.36
exclusion 30% noise	+5.9	−2.25	**0.038**	**0.041**
cancellation one-sided	+3.0	−1.06	0.31	0.31
cancellation partial coverage	+5.2	−2.04	0.057	0.059

Table 5: Peak-epoch comparisons against baseline ($n = 10$ per condition). Two-sided tests; Mann–Whitney is the non-parametric counterpart.